TECHNOLOGY STACK ANALYSIS

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine

1. Core Technology Selection

The technology stack for OptiCrop was selected to support fast execution, simple deployment, and data visualization:

Technology Component	Specific Choice	Role in Project
Programming Language	Python 3.10+	Data manipulation and model inference.
Web Framework	Flask 3.0.0	Maintains routes, handles forms, and renders template
Machine Learning	Scikit-Learn	StandardScaler, RandomForest, and KMeans model p
Data Structures	Pandas & NumPy	Data frame cleaning, formatting, and numeric operatio
Graphics Engine	Matplotlib & Seaborn	Generates performance evaluation plots.
Frontend CSS	Bootstrap 5 & Vanilla CSS	Styles the dark-themed glassmorphic user dashboard.

2. Design Architecture Rationale

Flask was chosen as the web frame because it integrates easily with Python-based scikit-learn models. A local CSV and SQLite database storage layout was selected to avoid external dependencies, keeping the application lightweight and simple to run.